1. Suppose we do Iterative Deepening Search (IDS) but instead of increasing depths as 1,2,3,4… we increase it as 1,2,4,8,16… i.e. double the depth instead of increasing it by 1. This modified version of IDS is
    a. Optimal but not complete
    b. Complete but not optimal
    c. More efficient than standard IDS in the worst case
    d. Less efficient than standard IDS in the worst case

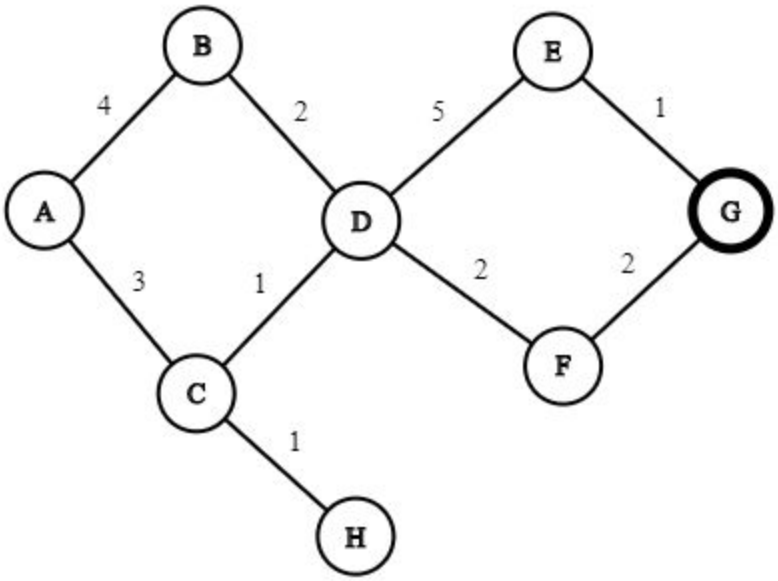
    The algorithm is complete because if a solution exists say at depth k the algorithm will surely explore it after (logk+1) rounds. The algorithm is not optimal because suppose there lies a solution at depth 2^k+1 in the rightmost node and there is some other non-optimal solution at a 2^{k+1} than the algorithm will not be able to find the optimal solution (assuming the algorithm stops after finding one solution).

    Suppose the goal lies at depth 9 in the rightmost node. Now IDS would not find solution till depth 8 and then on increasing depth till 16 it will expand all nodes upto depth 16 from left to right. Overall this would lead to more expanded nodes. In the usual case number of nodes expanded would have been close to 2^10 whereas in the modified IDFS it will be close to 2^16. So it is not efficient than standard IDS.

2. Suppose there is only one goal state and each step cost is k (k>0). Which of the following search algorithm(s) will return the optimal path?
    a. Breadth-First Search
    b. Depth First Search
    c. Uniform Cost Search
    d. Iterative Deepening Search

    If each step cost is the same than BFS and IDS will find the optimal solution. Uniform cost search will find optimal solution always. Depth First Search returns any one solution not necessarily optimal.

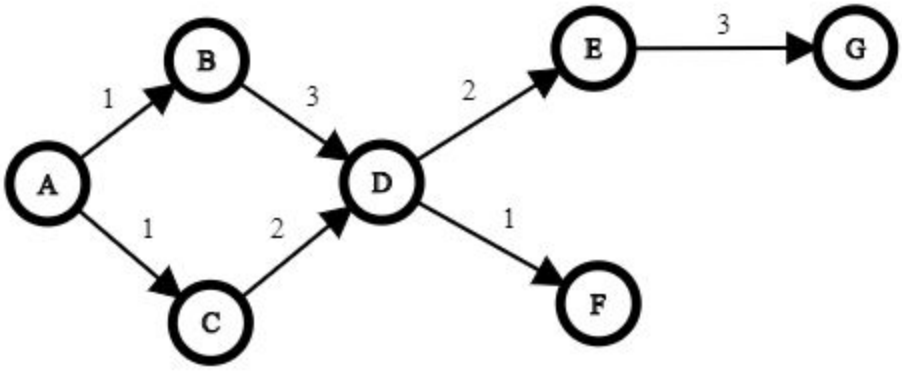
3. Consider the following graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Find the path to the goal found by Depth First Search with full duplicate detection, which explores children in lexicographical order. (Write answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD)



Ans. ABDEG

Since DFS is exploring in order lexicographic order it will explore nodes as
A -> B-> D-> C-> H-> backtrack to C -> backtrack to D -> E -> G

4. Consider the following directed graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Find the sequence of nodes explored (i.e., the sequence of removals from its frontier) by Uniform Cost Search, with no duplicate detection. Assume that, in case of ties, the search procedure uses a lexicographical order for tie-breaking. Output the order of nodes explored. (Write answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD)



Ans. ABCDDFEFEG

The key here was the lack of duplicate detection. So we can go back to same node multiple times. Rest of the algorithm is same as usual uniform cost search.

Iteration 1: A (Cost=0)
Iteration 2: B C (Cost=1)
Iteration 3: D (from C) (Cost=3)
Iteration 4: D (from B) F (From C,D) (Cost=4)
Iteration 5: E (from C,D) F (From B,D) (Cost=5)
Iteration 6: E (from B,D) (Cost=6)
Iteration 7: G (from C,D,E) (Cost=8)

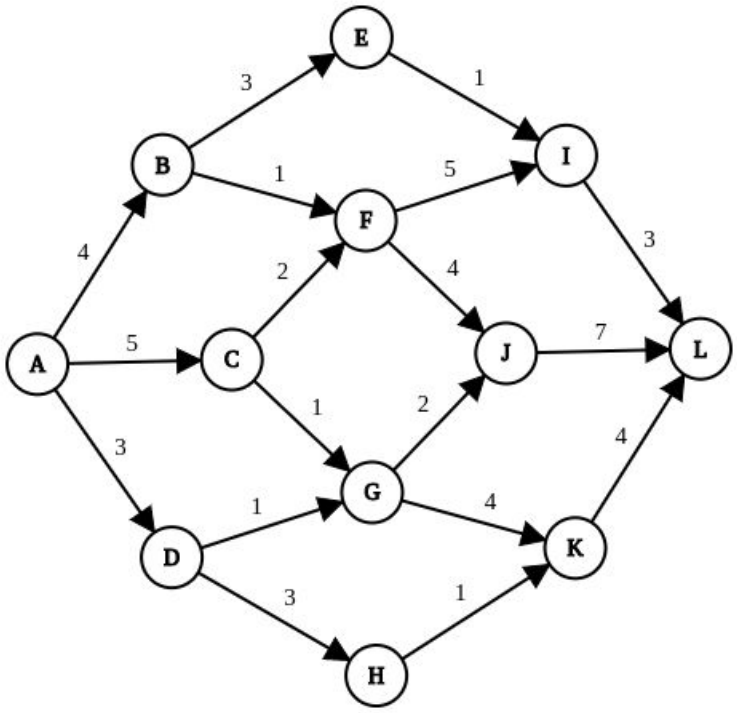
5. Which data structure is the most suited for implementing Uniform Cost Search?
    a. Queue
    b. Skip Lists
    c. Min Heap
    d. Max Heap

In Uniform Cost Search we need to search for the min cost node at each step so Min Heap is the most suited.

6. Which of the following uninformed search method(s) can be used to compute a solution to the N-Queens problem?
    a. Bi-Directional Search
    b. Beam Search
    c. Iterative Deepening Search
    d. Breadth-First Search

Bi-directional search cannot be used because the goal state is unknown. Beam search might not necessarily lead to a goal state. Iterative deepening search and an breadth first search are optimal (though inefficient).

7. Consider the following graph in which we are searching from start state A to goal state L. The number over each edge is the transition cost. Find the path to the goal found by Beam Search with a beam size of 2. The best partial solutions are the ones with the least cost. (Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD). Note: Conflicts are resolved by choosing a lexicographically shorter path if both have the same cost.



Ans: ADGKL

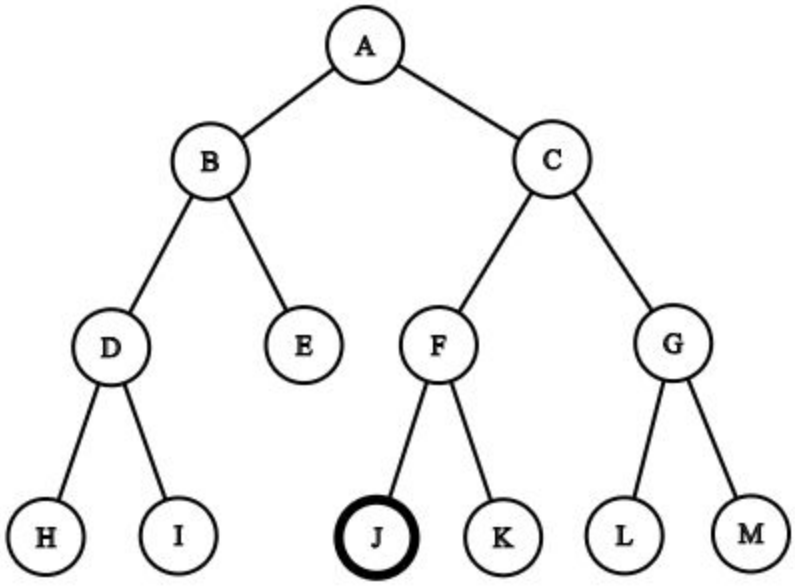
The paths in the square brackets represent the beam.
[A] -> [AB, AD] (AC is rejected) -> [ABF, ADG] (ABE, ADH are rejected) -> [ADGJ, ADGK] (ABFI, ABFJ are rejected) -> [ADGJL, ADGKL] . ADGKL is chosen as it is shorter. Note that it is not the optimal path (ADHKL and ABEIL are shorter than the ADGKL found by beam search).

8. Consider a full binary tree (Every internal node has degree 2 and there are 2^n leaves) of height n. Enumerate the nodes in the level at height (n - 1) from 1 to 2^{n-1} (left to right) and let the goal state X be the (2^{n-2} + 1) th node at that level. What is the absolute difference between the number of nodes explored by depth-first search and breadth-first search algorithms starting at the root? Assume n > 1 and left-to-right exploration.
    a. 2^{n-2} + n - 1
    b. 2^{n-1}
    c. 2n - 2
    d. 2^n - 2n + 2

No. of nodes explored by DFS = 1 (root) + (2^n - 1) (left-subtree) + (n - 1) (trail to the goal including the goal) = 2^n + n - 1 = 4.2^{n-2} + n - 1.
No. of nodes explored by BFS = (2^{n-1} - 1) (full tree upto height n-2) + 2^{n-2} (nodes to the left of X in level at height n-1) + 1 (X) = 2^{n-1} + 2^{n-2} = 3.2^{n-2}.

Difference = 2^{n-2} + n - 1

9. Consider the following graph in which we are searching from start state A to the goal state J. What is the number of nodes generated by Iterative Deepening Search? Assume that nodes are explored in right-to-left order and all edge costs are 1. Note: Exploration of the goal state J should be added to the count.



Ans: 19

First iteration = A
Second iteration = A, C, B
Third iteration = A, C, G, F, B, E, D
Fourth iteration = A, C, G, M, L, F, K, J
Total = 1 + 3 + 7 + 8 = 19

10. Consider the depth-limited search algorithm with depth limit f. If the only goal state exists at depth d then which of the following statement(s) is/are correct?
    a. **If f > d then it is complete but not optimal.**
    b. If f > d then it is not complete and not optimal.
    c. If f < d then it is complete but not optimal
    d. **If f < d then it is not complete and not optimal.**

If f > d, then the goal will be visited at some point hence the algorithm is complete. But it is clearly not optimal like depth first search. If f < d, then the goal will never be visited and hence the algorithm is not even complete.